



### Useful Formulas for kinematics

$$(1) v = v_0 + at$$

$$(2) \Delta x = v_0 t + \frac{1}{2}at^2$$

$$(3) v^2 = v_0^2 + 2a\Delta x$$

Q1: You are standing on a bridge that is 80 m high. You throw a rock perfectly horizontally at a speed of 15 m/s. We use  $g = -10 \text{ m/s}^2$ .

- (a) What is its initial vertical velocity?
- (b) Calculate the time it takes to hit the water below.
- (c) Calculate the horizontal distance the rock traveled before hitting the water.

Q2: A snowboarder wants to slide off a perfectly flat horizontal ledge that is 45 m high. To win the competition, he must safely land on a specific snow pad that is exactly 60m horizontally away from the base of the ledge.

- (a) Using the height of the ledge, calculate how long (t) the snowboarder will be in the air.
- (b) What must be the snowboarder's initial horizontal speed as he leaves the ledge to land exactly on the target?

Q3: A golfer hits a golf ball with an initial velocity of 50m/s at an angle of  $37^\circ$  above the ground.

- (a) Calculate the initial horizontal velocity ( $v_{0x}$ ) and vertical velocity ( $v_{0y}$ ).
- (b) Calculate the maximum height (H) the golf ball reaches.
- (c) Calculate the time that the golf ball takes to reach the highest point.

Q4: Based on the golf ball from Q3:

- (a) Find the total time the ball is in the air.

(b) Calculate the total horizontal distance the golf ball traveled.

Q5: Based on the exact same golf ball from Q3 and Q4, and use the symmetry of projectiles:

(a) Right before it hits the ground, what is its horizontal velocity ( $v_x$ ) and vertical velocity ( $v_y$ ) respectively?

(b) Using your answers in (a) and the Pythagorean theorem, calculate the total speed of the ball right before it hits the ground.

Q6: A catapult fires a heavy stone with an initial velocity of 50 m/s at an angle of  $53^\circ$  above the ground. There is an enemy castle wall that is exactly 120 m away horizontally. The height of this castle wall is 60 m.

(a) What is the horizontal and vertical velocity  $v_{0x}$  and  $v_{0y}$ ?

(b) Calculate how many seconds (t) it takes for the stone to travel 120 m horizontally.

(c) Calculate the height of the stone at that exact moment.

(d) Did the stone fly over the wall? Explain your reason.